

Deep Learning for Natural Language Processing

Assignment 3 — Solutions

Week 3: Word Representations & Cross-Lingual Representations

Total: 7 Questions | 10 Marks

1. Match each situation to the most appropriate family of cross-lingual embedding model (each family used at most once). [2 Marks]

- (a) Large monolingual corpora for both languages plus a dictionary of about 5,000 translated frequent words, but no parallel sentences.
- (b) Only topic-aligned encyclopaedia articles: no dictionary, no sentence alignments, no word alignments.
- (c) A moderate sentence-aligned parallel corpus without word alignments, plus a very large monolingual corpus for the high-resource language only.

A) (a) monolingual mapping, (b) pseudo-cross-lingual training, (c) joint optimization.

B) (a) joint optimization, (b) cross-lingual training, (c) monolingual mapping.

C) (a) pseudo-cross-lingual training, (b) monolingual mapping, (c) cross-lingual training.

D) (a) cross-lingual training, (b) joint optimization, (c) pseudo-cross-lingual training.

Answer & Solution

Answer: A

Solution: The lecture defines the four families and orders the supervision types from most to least expensive (word-aligned > sentence-aligned > document-aligned > lexicon > none).

- **(a) Monolingual mapping** ✓ “Train monolingual word embeddings on large monolingual corpora. Then learn a linear mapping.” The lecture uses exactly this setting — translate the 5,000 most frequent words, use those pairs as a bilingual dictionary, and learn W by $\min_W \sum_{i=1}^n \|W\mathbf{x}_i - \mathbf{z}_i\|^2$ with SGD.
- **(b) Pseudo-cross-lingual** ✓ “Create a pseudo-cross-lingual corpus by mixing contexts of different languages.” For document-aligned data the recipe is merge-and-shuffle: concatenate the aligned documents and randomly permute the words, which yields bilingual contexts for every word. No dictionary or sentence alignment required.
- **(c) Joint optimization** ✓ “Train their models on parallel (and optionally monolingual data). They jointly optimise a combination of monolingual and cross-lingual losses.” BilBOWA is the exact match: it needs no word alignments because it minimises the distance between the *means* of the word representations in aligned sentences, and it “leverages additional monolingual data”.

Cross-lingual training is the family deliberately left out: it optimises a cross-lingual constraint on a parallel corpus but does not add a monolingual objective.

[Reference: Lecture 15, Slides 112–113 (Types of models; Dataset Requirements); Slide 116 (linear mapping); Slide 119 (Merge and Shuffle); Slide 125 (BilBOWA)]

2. A Skip-gram model with full softmax has two-dimensional embeddings and vocabulary $V = \{\text{river, delta, silt, ledger, audit}\}$. The target (centre) vector of *river* is $\mathbf{v}_{\text{river}} = (2, 1)$ and the context vectors are

$$\begin{aligned}\mathbf{u}_{\text{river}} &= (0.5, -1), & \mathbf{u}_{\text{delta}} &= (1, 0), & \mathbf{u}_{\text{silt}} &= (0, 1), \\ \mathbf{u}_{\text{ledger}} &= (-1, 1), & \mathbf{u}_{\text{audit}} &= (-0.5, 0),\end{aligned}$$

with $P(o | c) = \exp(\mathbf{u}_o^\top \mathbf{v}_c) / \sum_{w \in V} \exp(\mathbf{u}_w^\top \mathbf{v}_c)$. Use $e^0 = 1$, $e^1 = 2.718$, $e^2 = 7.389$, $e^{-1} = 0.368$. With *river* as the centre word, what are $P(\text{delta} | \text{river})$, and $P(\text{silt} | \text{river})$? **[1 Mark]**

- A) 0.240 and 0.510
B) 0.500 and 0.250
C) 0.230 and 0.624
D) 0.624 and 0.230

Answer & Solution

Answer: D

Solution:

Step 1 — scores $\mathbf{u}_w^\top \mathbf{v}_{\text{river}}$ with $\mathbf{v} = (2, 1)$:

Word	Score	Word	Score
river	$0.5(2) + (-1)(1) = 0$	ledger	$-1(2) + 1(1) = -1$
delta	$1(2) + 0(1) = 2$	audit	$-0.5(2) + 0(1) = -1$
silt	$0(2) + 1(1) = 1$		

Step 2 — denominator: $\sum_w e^{\text{score}} = e^0 + e^2 + e^1 + e^{-1} + e^{-1} = 1 + 7.389 + 2.718 + 0.368 + 0.368 = 11.843$.

Step 3 — probabilities:

$$P(\text{delta} | \text{river}) = \frac{7.389}{11.843} = \mathbf{0.624}, \quad P(\text{silt} | \text{river}) = \frac{2.718}{11.843} = \mathbf{0.230}.$$

[Reference: Lecture 12, Slides 36–37 (Word2Vec Objective; Understanding $P(o|c)$); repeated at Slides 45–46]

3. A corpus consists of exactly two sentences: “*solar panels charge the battery*” and “*wind turbines spin*”. Skip-gram pairs use a window of 2 words on each side of the target, truncated at sentence boundaries (windows never cross between sentences). How many (target, context) pairs are produced in total, and how many have *charge* as the target? **[1 Mark]**

- A) 32 pairs; 4 with *charge* as target.
B) 20 pairs; 4 with *charge* as target.
C) 20 pairs; 2 with *charge* as target.
D) 14 pairs; 4 with *charge* as target.

Answer & Solution

Answer: B

Solution: Count the available context positions for each target.

Sentence	Positions available per target	Total
<i>solar</i> ₁ <i>panels</i> ₂ <i>charge</i> ₃ <i>the</i> ₄ <i>battery</i> ₅	2, 3, 4, 3, 2	14
<i>wind</i> ₁ <i>turbines</i> ₂ <i>spin</i> ₃	2, 2, 2	6
Grand total		20

charge sits at position 3 with a complete window on both sides, giving 4 pairs: (charge, solar), (charge, panels), (charge, the), (charge, battery).

Option A is the trap: $2mL = 2(2)(5) + 2(2)(3) = 20 + 12 = 32$ assumes every token has a full window, but truncation at the four sentence edges removes 12 pairs.

[Reference: Lecture 12, Slides 34–35 (example windows); Lecture 13, Slide 69 (Skip-Gram Training data)]

4. A linear mapping W projects source-language embeddings into the target-language space, obtained by minimising $\sum_i \|W\mathbf{x}_i - \mathbf{z}_i\|^2$ over a bilingual dictionary. With

$$W = \begin{pmatrix} 0.5 & 1 \\ 1 & -0.5 \end{pmatrix}, \quad \mathbf{x}_{\text{libro}} = (2, 1)^\top,$$

consider the following three candidate target-language vectors:

$$\mathbf{z}_{\text{book}} = (1.8, 1.6)^\top, \quad \mathbf{z}_{\text{bridge}} = (2.5, 0.5)^\top, \quad \mathbf{z}_{\text{house}} = (1.5, 1.0)^\top.$$

Which option correctly gives the mapped vector, the squared Euclidean distance to each candidate, and the nearest target-language vector? **[2 Marks]**

- A) $W\mathbf{x} = (2, 1.5)$; $d_{\text{book}}^2 = 1.25$, $d_{\text{bridge}}^2 = 0.05$, $d_{\text{house}}^2 = 0.50 \Rightarrow \text{bridge}$.
 B) $W\mathbf{x} = (2, 1.5)$; $d_{\text{book}}^2 = 0.05$, $d_{\text{bridge}}^2 = 1.25$, $d_{\text{house}}^2 = 0.50 \Rightarrow \text{book}$.
 C) $W\mathbf{x} = (1, -0.5)$; $d_{\text{book}}^2 = 5.05$, $d_{\text{bridge}}^2 = 3.25$, $d_{\text{house}}^2 = 2.50 \Rightarrow \text{house}$.
 D) $W\mathbf{x} = (2, 1.5)$; $d_{\text{book}}^2 = 0.50$, $d_{\text{bridge}}^2 = 1.25$, $d_{\text{house}}^2 = 0.05 \Rightarrow \text{house}$.

Answer & Solution

Answer: B

Solution:

Step 1 — apply the mapping.

$$W\mathbf{x} = \begin{pmatrix} 0.5 & 1 \\ 1 & -0.5 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.5(2) + 1(1) \\ 1(2) + (-0.5)(1) \end{pmatrix} = \begin{pmatrix} 2 \\ 1.5 \end{pmatrix}.$$

Step 2 — compute all three squared distances.

To *book*:

$$W\mathbf{x} - \mathbf{z}_{\text{book}} = (2 - 1.8, 1.5 - 1.6) = (0.2, -0.1),$$

so

$$d_{\text{book}}^2 = (0.2)^2 + (-0.1)^2 = 0.04 + 0.01 = \mathbf{0.05}.$$

To *bridge*:

$$W\mathbf{x} - \mathbf{z}_{\text{bridge}} = (2 - 2.5, 1.5 - 0.5) = (-0.5, 1.0),$$

so

$$d_{\text{bridge}}^2 = (-0.5)^2 + (1.0)^2 = 0.25 + 1.00 = \mathbf{1.25}.$$

To *house*:

$$W\mathbf{x} - \mathbf{z}_{\text{house}} = (2 - 1.5, 1.5 - 1.0) = (0.5, 0.5),$$

so

$$d_{\text{house}}^2 = (0.5)^2 + (0.5)^2 = 0.25 + 0.25 = \mathbf{0.50}.$$

Therefore,

$$d_{\text{book}}^2 < d_{\text{house}}^2 < d_{\text{bridge}}^2,$$

and the mapped source vector is closest to *book*.

Option A swaps the distances to *book* and *bridge*. Option C applies the mapping incorrectly.

Option D assigns the smallest distance to the wrong candidate.

[Reference: Lecture 15, Slide 116 ($\min_W \sum_i \|Wx_i - z_i\|^2$); Slide 112, item 1 (Monolingual mapping)]

5. A bilingual model trains on aligned sentence pairs by minimising the squared Euclidean distance between the *means* of their word vectors. A source sentence has three words with vectors (1, 2), (3, 0), (2, 1); the aligned target sentence has two words with vectors (1, 2) and (3, 1). What are the two means and the loss? **[1 Mark]**

- A) Means (2, 1) and (2, 1.5); loss = 0.5.
B) Means (2, 1) and (4, 3); loss = 8.0.
C) Means (2, 1) and (2, 1.5); loss = 0.25.
D) Means (6, 3) and (4, 3); loss = 4.0.

Answer & Solution

Answer: C

Solution:

Step 1 — source mean. $\left(\frac{1+3+2}{3}, \frac{2+0+1}{3}\right) = \left(\frac{6}{3}, \frac{3}{3}\right) = \mathbf{(2, 1)}.$

Step 2 — target mean. $\left(\frac{1+3}{2}, \frac{2+1}{2}\right) = \mathbf{(2, 1.5)}.$

Step 3 — loss. Difference (0, -0.5), so

$$\|\cdot\|^2 = 0^2 + (-0.5)^2 = \mathbf{0.25}.$$

A training step adjusts the word vectors of both sentences so that the two means move together. The lecture makes the point that matters: “Instead of minimising the distance between words that were aligned to each other, they minimise the distance between the means of the word representations in the aligned sentences” — which is why BilBOWA needs sentence alignment but no word alignment. Option A reports the Euclidean distance (0.5) rather than its square; options B and D use sums instead of means.

[Reference: Lecture 15, Slide 125 (BilBOWA); Slide 121 (Bilingual compositional sentence model); Lecture 14, Slide 97 (mean sentence representations)]

6. Skip-gram with negative sampling uses

$$L = -\log \sigma(\mathbf{u}_{\text{pos}}^\top \mathbf{v}_w) - \sum_{i=1}^k \log \sigma(-\mathbf{u}_{\text{neg}_i}^\top \mathbf{v}_w), \quad \sigma(x) = \frac{1}{1 + e^{-x}}.$$

Here $\mathbf{v}_{\text{harbour}} = (1, 2)$, $\mathbf{u}_{\text{wharf}} = (1, 0.5)$ (positive), $\mathbf{u}_{\text{recipe}} = (-1, 0)$ and $\mathbf{u}_{\text{sprint}} = (0, 0.5)$ (negatives). Logarithms are natural; $\sigma(2) = 0.881$, $\sigma(1) = 0.731$, $\sigma(-1) = 0.269$, $\log 0.881 = -0.127$, $\log 0.731 = -0.313$, $\log 0.269 = -1.313$. What are the three loss terms, the total, and the required update? **[2 Marks]**

- A) 0.127, 1.313, 0.313; $L = 1.753$; *recipe* contributes more, and the update must decrease $\mathbf{u}_{\text{recipe}}^\top \mathbf{v}_{\text{harbour}}$.
- B) 0.127, 0.313, 1.313; $L = 1.753$; *sprint* contributes more, and the update must *decrease* $\mathbf{u}_{\text{sprint}}^\top \mathbf{v}_{\text{harbour}}$ and *increase* $\mathbf{u}_{\text{wharf}}^\top \mathbf{v}_{\text{harbour}}$.
- C) 0.881, 0.731, 0.269; $L = 1.881$; *wharf* contributes most, since it is the positive example.
- D) 0.127, 0.313, 1.313; $L = 1.753$; *sprint* contributes more, and the update must *increase* $\mathbf{u}_{\text{sprint}}^\top \mathbf{v}_{\text{harbour}}$ so that the model can tell the two negatives apart.

Answer & Solution

Answer: B

Solution:

Step 1 — dot products with $\mathbf{v} = (1, 2)$: $\mathbf{u}_{\text{wharf}}^\top \mathbf{v} = 1(1) + 0.5(2) = 2$; $\mathbf{u}_{\text{recipe}}^\top \mathbf{v} = -1(1) + 0(2) = -1$; $\mathbf{u}_{\text{sprint}}^\top \mathbf{v} = 0(1) + 0.5(2) = 1$.

Step 2 — loss terms.

positive (wharf): $-\log \sigma(2) = -\log 0.881 = 0.127$,

negative (recipe): $-\log \sigma(-(-1)) = -\log \sigma(1) = -\log 0.731 = 0.313$,

negative (sprint): $-\log \sigma(-1) = -\log 0.269 = 1.313$.

Step 3 — total. $L = 0.127 + 0.313 + 1.313 = \mathbf{1.753}$.

Step 4 — interpretation. *Sprint* dominates the loss (1.313 versus 0.313) because its dot product with the target is *positive* (+1): the model currently treats a non-context word as a plausible context, which is exactly what the negative term penalises. *Recipe* already has a negative dot product, so $\sigma(-\mathbf{u}^\top \mathbf{v}) = \sigma(1) = 0.731$ is already large and its penalty is small. The lecture states the goal directly: “Maximize the emb. similarity of the real target-context word pairs . . . Maximize the emb. distance between target noncontext word pairs”. The step must therefore **push $\mathbf{u}_{\text{sprint}}$ and $\mathbf{v}_{\text{harbour}}$ apart** and **pull $\mathbf{u}_{\text{wharf}}$ and $\mathbf{v}_{\text{harbour}}$ together**.
[Reference: Lecture 13, Slides 67–68 (Negative Sampling: Objective function); Slides 71–73 (Skip-gram model; Skip-gram learning algorithm)]

7. A team builds sparse word vectors from a 40,000-token corpus of gardening articles. Each word is a vector over 8,000 context-word dimensions weighted by positive pointwise mutual information. The words **spade** and **shovel** each occur about 60 times and are near-synonyms, yet their vectors share only two non-zero dimensions and their cosine similarity is 0.04. What is the cause, and which remedy from the lecture addresses it? **[1 Marks]**

- A) The distributional hypothesis fails for synonyms; widen the context window

- B) PPMI over-weights frequent contexts; switch to raw co-occurrence counts
- C) The vectors are too sparse to share dimensions; learn short dense vectors
- D) Cosine is unreliable on sparse vectors; use the unnormalised dot product

Answer & Solution

Answer: C

Solution: The distributional hypothesis concerns the *distribution over contexts*, which is badly undersampled here. With ≈ 60 tokens and a small window, each word touches at most a few hundred context tokens out of 8,000 dimensions, so both vectors are more than 97% zero.

Cosine similarity, $\cos(\mathbf{v}, \mathbf{w}) = \frac{\sum_i v_i w_i}{\|\mathbf{v}\| \|\mathbf{w}\|}$, is driven entirely by *shared* non-zero dimensions. Two near-synonyms described with different-but-related neighbours (“digging” versus “excavating”) therefore score near zero. This is the argument made in the lecture: two words are synonyms “but are distinct dimensions . . . a word with one as a neighbour and a word with the other as a neighbour should be similar, but aren’t”.

Remedy: short dense vectors (SVD/LSA, or a learned word2vec/GloVe embedding). Each dimension becomes a latent factor shared by many context words, so related-but-distinct contexts collapse onto the same dimensions and the two words end up close. Widening the window (A) helps only marginally and blurs the kind of similarity learned; the problem is not PPMI (B) nor the metric (D).

[Reference: Lecture 11, Slides 19–21 (Sparse versus Dense Vectors); Slide 6 (Distributional Hypothesis); Slide 16 (PMI)]

— End of Assignment —